

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 3
Comparative Analysis

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Assessment Tool Weightage: 40%

Code of Conduct:

D.Suryakiran/192325057(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

QUESTIONS:

Total Marks:25

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification application

Ans:CNN vs Fully Connected Neural Network

A **Convolutional Neural Network (CNN)** is more suitable for X-ray image classification than a traditional Fully Connected Neural Network (FCNN) because CNNs are specifically designed to process spatial information in images.

Aspect	CNN	Fully Connected Neural Network
Architecture	Convolution → activation → pooling → fully connected layers	Mainly fully connected layers
Input handling	Preserves 2D/3D spatial structure	Image is usually flattened into a 1D vector
Feature extraction	Automatically extracts local features	Learns features without explicitly considering spatial locality
Parameters	Fewer due to parameter sharing	Very large number of parameters
Computation	More efficient for images	Computationally expensive for high-resolution images
Spatial information	Preserved through convolution	Mostly lost after flattening
Translation handling	Better at recognizing features at different locations	Poorer location-based feature handling
Image suitability	Highly suitable	Less suitable for large images

Role of CNN Components

1. Convolutional layers:

These layers apply small filters to the image and detect local patterns such as edges, textures, shapes, and abnormalities.

2. Filters:

A filter slides across the image and produces a feature map. Different filters learn different visual characteristics.

3. Pooling layers:

Pooling reduces the spatial dimensions of feature maps, decreasing computation and making the network less sensitive to small changes in object location.

For example, in an X-ray image, early layers can detect **edges and textures**, while deeper layers can identify **bones, organs, lesions, or abnormal regions**.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of convolutional layers, pooling layers, activation layers, and fully connected layers in terms of feature extraction, spatial information, computational requirements, and classification. Further compare early-layer filters and deeper-layer filters based on the type of features they learn, and justify how these layers collectively improve defect detection.

Ans: Different CNN Layers and Filters

A CNN consists of several types of layers that work together to detect defects in industrial products.

Layer	Main Function	Spatial Information	Computational Requirement	Role in Classification
Convolutional	Extracts features	Preserves spatial relationships	Moderate	Detects defect patterns
Pooling	Reduces feature-map size	Partially reduces spatial information	Low	Makes features more robust
Activation	Adds non-linearity	Maintains dimensions	Low	Enables complex feature learning
Fully Connected	Combines extracted features	Spatial structure mostly removed	High	Produces final classification

1. Convolutional Layer

The convolutional layer applies filters to the input image. It identifies features such as:

- Edges
- Corners

- Textures
- Cracks
- Scratches
- Shape abnormalities

2. Pooling Layer

Pooling reduces the dimensions of feature maps.

For example, **Max Pooling** selects the maximum value from a small region. This reduces computation and helps the CNN recognize a feature even if its exact location changes slightly.

3. Activation Layer

Functions such as **ReLU** introduce non-linearity:

This allows the network to learn complex relationships rather than only simple linear patterns.

4. Fully Connected Layer

The extracted features are converted into classification decisions. For example:

Input → CNN → Features → Fully Connected Layer → Defective / Non-defective

Early vs Deep Filters

Early-layer filters:

- Detect simple edges
- Lines
- Corners
- Basic textures
- Color/intensity changes

Deeper-layer filters:

- Combine simple features
- Detect complex shapes
- Identify object parts
- Recognize complete defect patterns

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare parameter sharing in CNNs with independent weights in fully connected networks in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for largescale image-processing applications.

Ans: Parameter Sharing vs Independent Weights

Parameter sharing is one of the major advantages of CNNs for large-scale image processing.

Aspect	CNN Parameter Sharing	Fully Connected Independent Weights
Weights	Same filter weights reused across image	Different weights for each connection
Parameters	Much fewer	Very large
Memory	Lower	Higher
Computation	More efficient	Computationally expensive
Feature detection	Same feature can be detected anywhere	Feature detection depends strongly on learned connections
Translation handling	Better	Poorer
Learning efficiency	High	Lower for large images

Parameter Sharing

Suppose a CNN has a **3×3 filter**. The same nine weights are applied repeatedly across different locations of the image.

For example:

Filter → left region → center region → right region → other regions

The filter can detect the same feature wherever it appears.

In contrast, a fully connected network assigns separate weights to connections between every input pixel and neuron.

For an image of size:

there are:

input values.

A fully connected layer with 1,000 neurons would require approximately:

weights, excluding biases.

A CNN can use a much smaller number of convolutional parameters because the same filters are reused.

Translation-Related Invariance

If a defect moves from one part of an image to another, the same convolutional filter can detect it because the filter is applied throughout the image. Pooling further helps reduce sensitivity to small spatial changes.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

AlexNet vs ResNet

AlexNet and ResNet are important CNN architectures, but ResNet is generally more suitable for complex modern image-recognition applications.

Aspect	AlexNet	ResNet
Introduction	2012	2015
Depth	8 learnable layers	Can have 18, 34, 50, 101, 152+ layers
Architecture	Sequential CNN	Residual/skip connections
Filters	Uses convolutional filters including large early filters	Primarily smaller convolutional filters in residual blocks
Parameters	Relatively high, especially for its depth	Varies by version; deeper models can have many parameters
Feature extraction	Good hierarchical features	More powerful and deeper feature extraction
Training difficulty	Easier than very deep networks	Skip connections make very deep networks easier to train
Vanishing gradient	Less severe because network is shallower	Skip connections strongly help address degradation/gradient problems
Computational cost	Lower for the original architecture	Can be higher depending on ResNet depth
Performance	Historically excellent	Generally better for complex image-recognition tasks

AlexNet

AlexNet uses convolutional layers, ReLU activation, pooling, and fully connected layers. It demonstrated the effectiveness of deep CNNs for large-scale image classification.

ResNet

ResNet introduces **residual/skip connections**. Instead of forcing every layer to learn a complete transformation, a residual block learns a residual function:

Here, x is the original input and represents the learned transformation.

The shortcut connection allows information and gradients to flow more easily through deep networks.

Vanishing-Gradient Problem

As networks become deeper, gradients can become extremely small during backpropagation. This makes training difficult.

ResNet's skip connections help gradients pass through the network more effectively, allowing much deeper models to be trained.

Recommendation

For a **complex image-recognition application**, **ResNet is recommended** because it:

- Supports much deeper architectures.
- Learns more complex hierarchical features.
- Reduces degradation and vanishing-gradient difficulties.
- Provides strong classification performance.
- Can be adapted to different computational requirements using ResNet-18, ResNet-34, ResNet-50, etc.

Therefore, **ResNet is generally a better choice than AlexNet for modern complex computer-vision applications**, although AlexNet can still be useful when simplicity and lower computational requirements are priorities.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Regularized CNN vs Non-Regularized CNN

The agricultural CNN has **high training accuracy but poor performance on unseen images**, which indicates **overfitting**. Regularization techniques should therefore be applied.

Aspect	Non-Regularized CNN	Regularized CNN
Overfitting	High	Reduced
Generalization	Poor	Better
Training stability	May be unstable	More stable
Training accuracy	Can become extremely high	Usually slightly lower
Validation accuracy	Often poor	Usually improved
Memory/computation	Lower	Slightly higher depending on technique
Real-world performance	Poorer	Better

1. Dropout

Dropout randomly deactivates a fraction of neurons during training.

For example, with a dropout rate of 0.5, approximately half of the selected neurons are ignored during a training step.

This prevents the network from depending too heavily on particular neurons and reduces overfitting.

2. Data Augmentation

Data augmentation creates modified versions of training images using transformations such as:

- Rotation
- Flipping
- Cropping
- Scaling
- Translation
- Brightness changes

For plant disease detection, this is especially useful because leaves may appear at different angles, sizes, orientations, and lighting conditions.

3. Batch Normalization

Batch normalization normalizes intermediate activations during training. It can:

- Improve training stability
- Allow faster training
- Improve gradient flow
- Reduce sensitivity to initialization

Recommended Approach

A suitable architecture would be:

Leaf Image → Data Augmentation → Convolution → ReLU → Batch Normalization → Pooling → Convolution → ReLU → Batch Normalization → Pooling → Dropout → Fully Connected → Disease Class

Best Combination

For this plant-disease application:

Data augmentation + Batch Normalization + Dropout

is a strong practical combination.

- **Data augmentation** improves the diversity of training data.
- **Batch normalization** improves training stability.
- **Dropout** reduces over-reliance on individual neurons.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies

Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand